

a strong preference for box-type test equipment. The main reason is that debugging of firmware and repair of box-type equipment is far easier than the module. Having said this, certain niche industries like automation are witnessing good demand of PXI/VXI instrumentation. However, the requirements of aerospace and defence and telecom industries are mostly met with box-based solutions only.

Of late, some PXI-based solutions have been introduced in the telecom market. It is too early to comment on their functionality as these devices are yet to be tested and accepted by users.

How to best manage T&M equipment?

Most T&M equipment are designed for lab usage, which requires a controlled operating temperature. The optimal performance of the equipment is specified in the range of 23 to 28°C, though most of the equipment are usable from 5 to 45°C. Regular calibration at a time interval recommended by the manufacturer (every one, two or three years) would ensure that the equipment performs within the specified limits.

What are the challenges faced while working with the new 802.11ac?

Today's smartphones carry both cellular and non-cellular technologies at the same time. The number of mobile applications has grown along with the increased use of smartphones and tablets, resulting in a huge rise in bandwidth demand. To a large extent, this demand is driven by the popularity of streaming videos on smartphones and tablets.

For cellular providers, this increased demand has burdened their network capacity, leaving them wanting more. Network operators are now slowly knocking off their unlimited usage plans and going back to metered plans. This increased demand is also taxing the Wi-Fi network to supply sufficient bandwidth, especially as there is a large number of powerful mobile handsets arriving on the market every day. Bulk file transfers, streaming videos, database searches and voice over Wi-Fi are placing increasing demands on a network's capability to provide consistent bandwidth. This is where technologies like 802.11ac are becoming increasingly important due to the large bandwidth and advanced MIMO techniques supported by them.

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challenges. 802.11ac devices make use of OFDM modulation, wider channel bandwidth than 802.11n, higher-order modulation with eight streams and enhanced multi-user MIMO techniques. So designers and manufacturers need to understand the complex requirements not only from product creation context but also from the way testing has to be carried out.

The testers will need wide IF bandwidths in vector signal analysers and generators (VSAs and VSGs) and improved transmitter modulation accuracy for testing multi-user MIMO requirements. Single-box testers combining a VSA and VSG of today will not be able to test 802.11ac due to the increased bandwidth requirements. More number of VSAs and VSGs will be required to test the eight streams as part of the multi-user MIMO technique. The challenge to accurately measure the transmitter modulation accuracy will mean that the VSA should have a very good EVM specification.

How are airborne, shipborne and tactical radio-communications different from hardware point of view?

The radio technology is the same for airborne, shipborne and tactical use, but the form factor changes as per the requirement of the platform. For example, manpack use or portability can be the requirement for tactical radios. Fitting them in a given fixed space is a requirement for shipborne or airborne use.

Secure radio communication is one of the key technology areas for R&S. There is a paradigm shift from hardware-centric radios to software-centric radios. R&S is actively and consistently driving technology innovations to achieve truly multiband, multirole and multimode radios for airborne, shipborne and tactical use. ●

(The complete interview can be viewed at electronicsforu.com)

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